

Chapter 8: Model Evaluation

Learning Objectives

By the end of this chapter, you will:

- Understand how AI models are evaluated before and after deployment.
 - Learn the key metrics used to measure AI system performance in enterprise environments.
 - Develop a framework for monitoring and improving AI products throughout their lifecycle.
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Beginner Explanation

Building an AI model is only half the job. The real question is:

"How do we know if the AI is actually doing a good job?"

This process is called **Model Evaluation**.

Think about hiring an employee. You would evaluate:

- Accuracy
- Productivity
- Reliability
- Communication
- Consistency

AI models are evaluated in a similar way.

Example

Suppose an AI assistant answers 100 employee questions.

Metric	Result
Correct Answers	92
Incorrect Answers	8
Average Response Time	3 seconds
User Satisfaction	89%

Based on these results, we can determine whether the model is performing well.

Common Evaluation Metrics

Metric	Meaning
Accuracy	How often the model is correct
Precision	How relevant responses are
Recall	How much relevant information is found
Latency	Time taken to respond
Cost	Cost per request
Reliability	Consistency over time

As an AI TPM, your responsibility is to ensure that AI systems continuously meet business expectations.

Intermediate Explanation

Enterprise AI systems require ongoing evaluations because performance can change over time due to:

- Model updates
- New data
- User behavior changes
- Infrastructure changes
- Cost fluctuations

Evaluation Categories

Category	Questions to Ask
Accuracy	Is the response correct?
Relevance	Is the answer useful?
Safety	Is the output appropriate?
Performance	Is the system fast enough?
Cost	Is the solution within budget?
Reliability	Is performance consistent?

Enterprise Evaluation Framework

1. Define success metrics.
2. Build evaluation datasets.
3. Execute automated tests.
4. Conduct human reviews.
5. Monitor production metrics.
6. Improve continuously.

Human-in-the-Loop Evaluation

Most enterprises use a combination of:

- Automated evaluations
 - Human reviews
 - User feedback
 - A/B testing
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Why This Matters for an AI TPM

Responsibility	Impact
Delivery	Ensures AI systems meet agreed-upon performance targets.
Risk	Identifies hallucinations, bias, and reliability issues early.
Stakeholder Management	Provides measurable KPIs to leadership teams.
Budget	Tracks operational efficiency and cost optimization.

Enterprise Example

Business Problem

A financial institution deployed an AI assistant to help employees answer compliance-related questions.

Initial feedback indicated inconsistent responses across departments.

Solution

Implement a Model Evaluation framework using Azure AI Foundry.

Architecture

- Azure OpenAI
- Azure AI Foundry Evaluations
- Azure AI Search
- Azure Monitor
- Power BI

Outcome

Metric	Before	After
Accuracy	82%	96%
User Satisfaction	76%	92%

Metric	Before	After
Latency	5 Seconds	3 Seconds
Hallucination Rate	14%	3%

Lessons Learned

- Evaluation should begin during development.
- Human reviewers remain essential.
- KPIs should be continuously monitored.
- Performance improvements often come from better prompts and retrieval.

Azure Implementation

Services Used

Service	Purpose
Azure OpenAI	Model inference
Azure AI Search	Context retrieval
Azure Functions	Evaluation orchestration
Azure AI Foundry	Automated evaluations

Implementation Workflow

Step 1: Define Metrics

Examples:

- Accuracy
- Latency
- Cost
- Reliability
- User Satisfaction

Step 2: Create Evaluation Dataset

Sample inputs:

Question
What is our PTO policy?
How do I submit expenses?

Question

What are the compliance requirements?

Expected outputs are maintained for comparison.

Step 3: Run Evaluations

Measure:

- Response quality
 - Hallucinations
 - Safety
 - Consistency
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Step 4: Human Validation

Evaluate:

- Relevance
 - Tone
 - Completeness
 - Business impact
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Step 5: Continuous Monitoring

Track:

- Daily accuracy
 - Monthly costs
 - Adoption rates
 - Error trends
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Mermaid Diagram

```
```mermaid id="6hy4zq" graph TD
```

```
A[User Queries]
```

```
A --> B[Azure OpenAI]
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B --> C[Azure AI Foundry Evaluations]
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C --> D[Performance Metrics]
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D --> E[Power BI Dashboard]

E --> F[AI TPM Review]

F --> G[Continuous Improvements] ``

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## Evaluation Scorecard

Metric	Target
Accuracy	>95%
Precision	>90%
Recall	>90%
Latency	<3 Seconds
Cost per Request	<\$0.05
User Satisfaction	>90%

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## TPM Checklist

- ☐ Define evaluation criteria.
- ☐ Create test datasets.
- ☐ Configure automated evaluations.
- ☐ Conduct human reviews.
- ☐ Establish KPI dashboards.
- ☐ Monitor hallucination rates.
- ☐ Track costs.
- ☐ Review stakeholder feedback.
- ☐ Document lessons learned.
- ☐ Schedule monthly evaluations.

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## Templates

### Model Evaluation Template

Metric	Current	Target
Accuracy		
Latency		
Reliability		
Cost		

Metric	Current	Target
Satisfaction		

## Monthly Review Dashboard

KPI	Status
Accuracy	Green
Costs	Green
Adoption	Yellow
Reliability	Green
Risks	Yellow

## Risk Register

Risk	Impact	Mitigation
Hallucinations	High	Human review
High Latency	Medium	Infrastructure optimization
Cost Overruns	Medium	Budget alerts
Low Adoption	Medium	User training

## Common Mistakes

1. Evaluating only during development.
2. Ignoring user feedback.
3. Focusing solely on accuracy.
4. Failing to monitor costs.
5. Neglecting hallucination tracking.
6. Not involving business stakeholders.
7. Assuming performance remains constant over time.

## Key Takeaways

- Model Evaluation is a continuous process, not a one-time activity.
- Successful AI programs require measurable KPIs and regular assessments.
- Azure AI Foundry provides robust evaluation capabilities for enterprise AI systems.
- Human reviews remain a critical component of AI quality assurance.
- AI TPMs must balance performance, reliability, cost, and business outcomes.

- Continuous improvement is essential for long-term AI success.

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**Footer**

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